Retinal Vessel Tree-based Identification by Feature Matching

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Objective of the Project

This project aims to develop a sophisticated biometric identification system that utilizes retinal vessel tree analysis.

Our proposed solution is a splitted in 3 different tasks:

- Optic disc identification
- Vessel tree extraction
- Vessel tree matching

Why do the analyses?

Early diagnosis of eye and systemic diseases

The identification and segmentation of the retinal vascular tree is central to detecting diseases such as diabetic retinopathy, glaucoma, age-related macular degeneration, hypertension, and other cardiovascular diseases. Alterations in the retinal vascular structure, such as tortuosity or narrowing of the vessels, are early indicators of many of these disorders

Disease research and modelling

The data extracted from the retinal vascular tree identification is used to study the physiology of the retina and its interactions with systemic diseases. These studies also help in the development of new treatments and personalised therapies

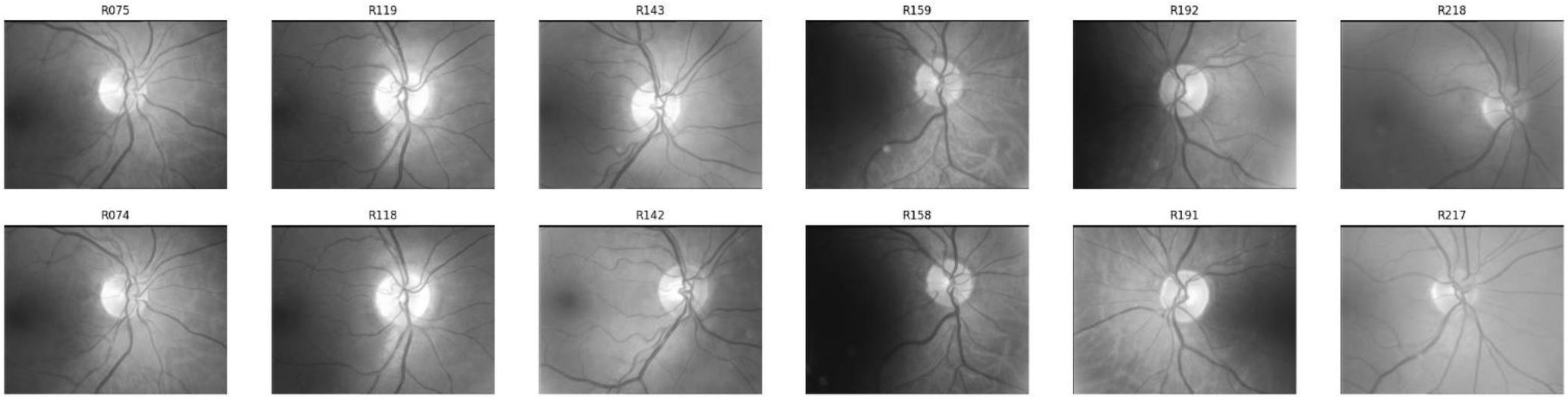
Monitoring of chronic diseases

This process helps to assess the evolution of chronic diseases such as diabetes or hypertension. Changes in retinal vessels (dilation, bleeding, etc.) provide clinical information to adapt therapies and improve patients' quality of life

Analysis of vascular structure

The retinal vascular tree is a highly branched and complex structure. Measuring parameters such as length, diameter and tortuosity of the vessels supports the recognition of anatomical or functional abnormalities that may result from genetic diseases, ischemia or trauma.

Dataset



The dataset contains 6 test images and 6 reference images of the same retinal vessel trees, with variations in perspective, translation, rotation, and intensity.

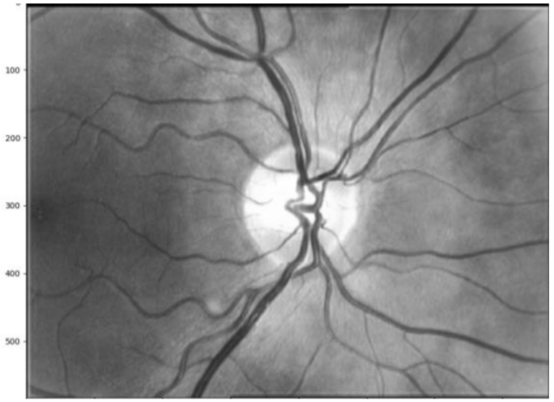
These variations introduce challenges such as noise, uneven illumination, and misalignment, making the dataset ideal for testing robustness.

Our technical Approach

Preprocessing

Preprocessing steps

1. **Contrast Enhancement with CLAHE:** helps to enhance the visibility of vessel structures and image contrast across all regions.
2. **Noise Reduction with Gaussian Blurring:** This improves the reliability of subsequent vessel segmentation techniques, especially thresholding.

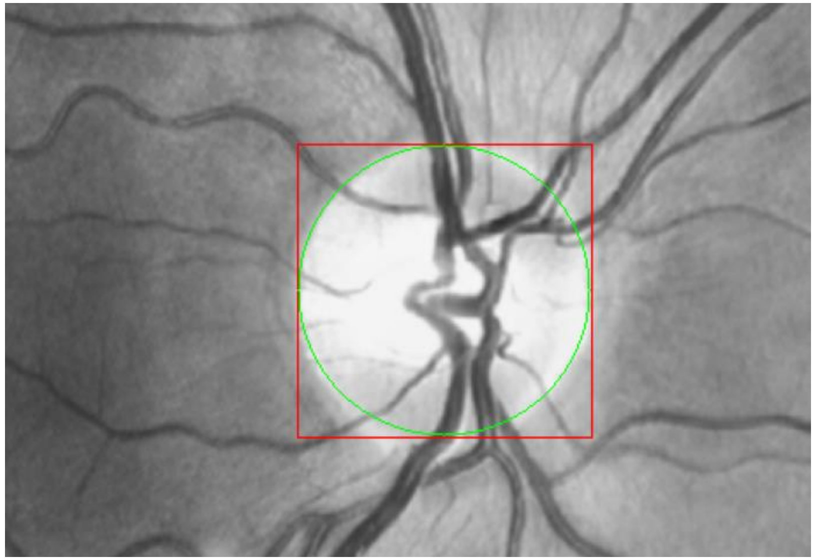


Our technical Approach

Preprocessing → Optic Disc Location

Opting Disk Finding steps

1. **Template Definition:** A circular template is generated to represent the optic disc, leveraging its consistent dimensions across various retinal images.
2. **Template Matching** using the synthetic template and TM_CCORR/TM_SQDIFF/TM_CCOEFF method
3. **Best Match Identification:** Locate center and radius using cv2.minMaxLoc



Our technical Approach

Preprocessing → Optic Disc Location → Vessel Tree Extraction

Mask creation

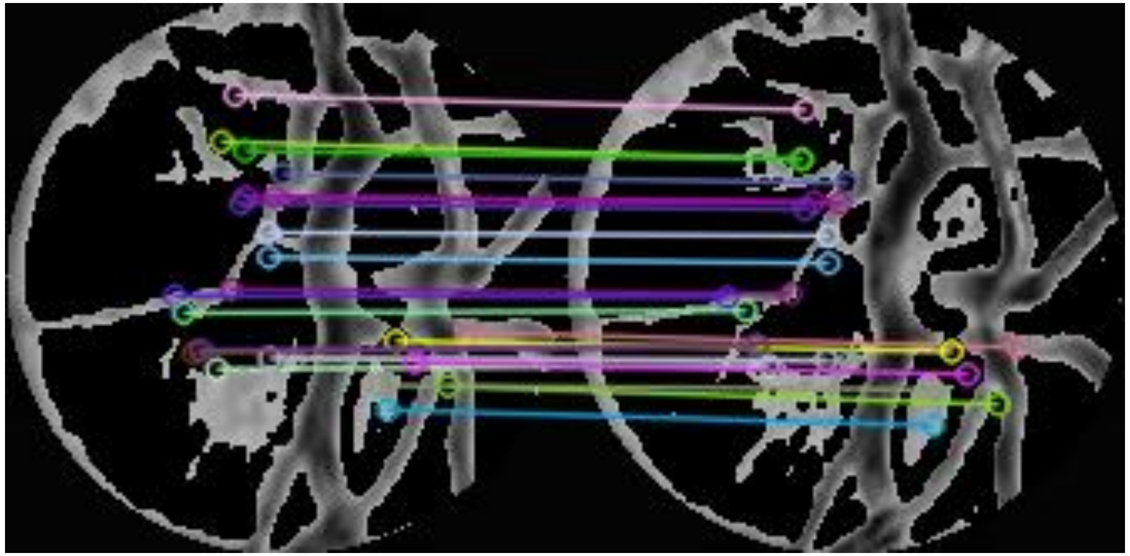
1. **Mask creation:** Thresholding the optic disc for generating vessel tree mask. Apply morph erode for smooth edges
2. **Final extraction:** Extract the vessel tree by applying the mask with cv2.bitwise_and

Our technical Approach

Preprocessing → Optic Disc Location → Vessel Tree Extraction → Vessel Tree Matching

Vessel Tree Matching Pipeline

1. **Keypoint extraction:** AKAZE algorithm is used to detect keypoints and generate keypoints
2. **Keypoints Brute-force Matching:** Hamming distance measures the similarity between descriptors, filtering out incorrect matches.
3. **Homography Estimation:** applying RANSAC, for aligning and verifying the match between two images. Useful for cleaning outliers in the matching process and refines the accuracy of the comparison.



Results

	R074	R118	R142	R158	R191	R217
R075	0.96	0	0	0	0	0
R119	0	0.90	0	0	0	0
R143	0	0	1	0	0	0
R159	0	0	0	1	0	0
R192	0	0	0	0	0.60	0
R218	0	0	0	0	0	0.91

Different trees

When comparing descriptors from dissimilar vessel trees, our pipeline finds no shared features, resulting in a similarity score of 0.

Similar trees

The majority of matches (85% or higher) exhibit high similarity scores greater than 85%, highlighting the effectiveness of our approach in identifying matching patterns

Notable exceptions

The R192-R191 match demonstrates a slightly lower similarity with a score of 60%.

Conclusion and future works

This project leverages computer vision algorithms to achieve high performance. However, we recognize that achieving optimal results with larger datasets can be challenging due to factors like noise, uneven intensity distribution, etc.

Proposed future solutions:

- Explore other computer vision algorithms
- Try AI based solutions (SAM + Embedding + cousine similarity)
- Test solution on larger datasets